

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think> in the context of human immunodeficiency virus (HIV), the CD21 molecule plays a role in the virus's entry into the host cells </think><answer> the pathogenesis of human immunodeficiency virus (HIV) would most likely be affected by this genetic modification </answer>

